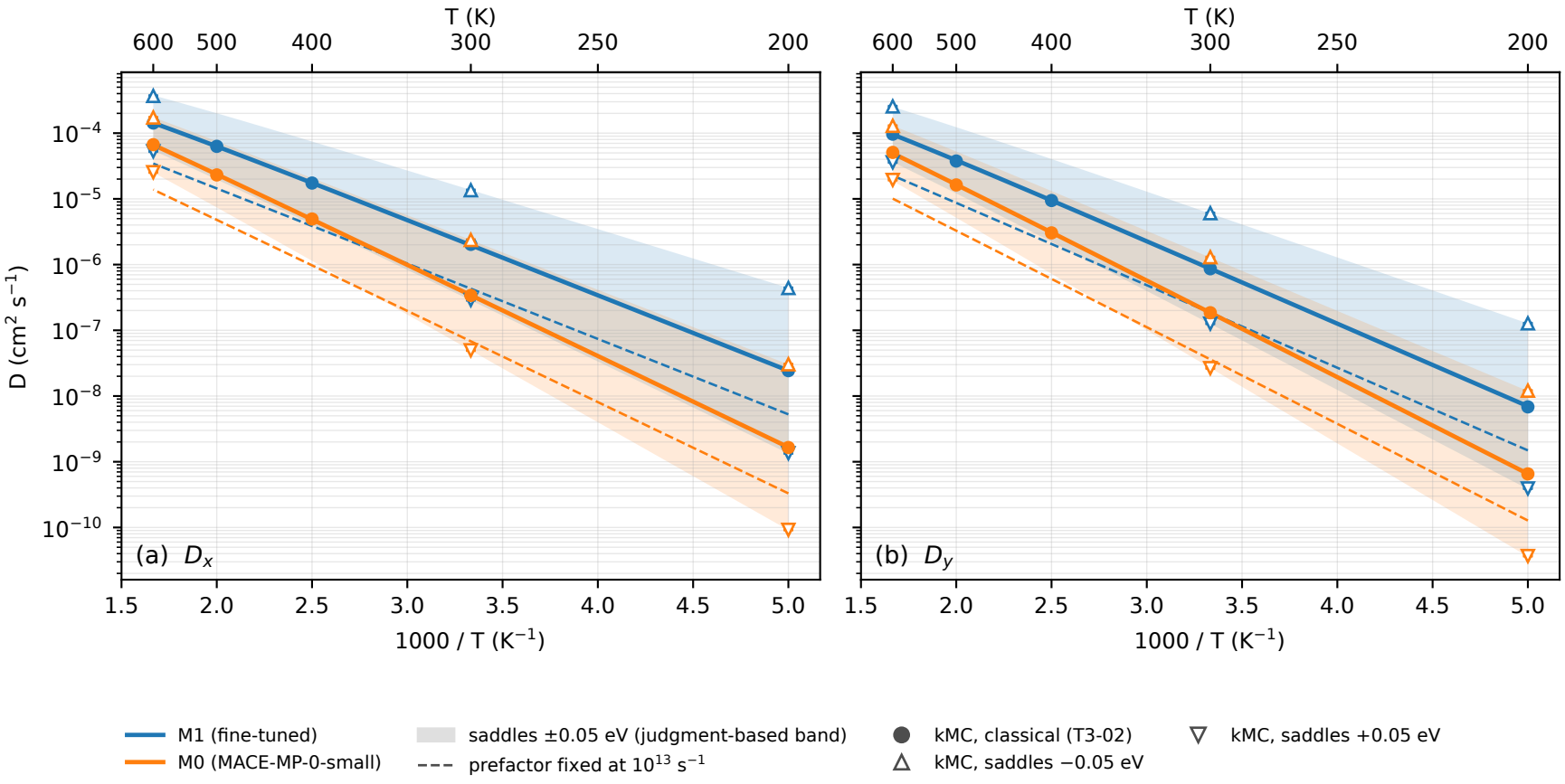
H on Nb-terminated NbP (1 ML barriers): lattice kMC / exact D, MLIP-based hTST (not DFT)
T (K)
600
500
400
300
250
200
D (cm^2 s^{-1})
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
(a) D_x
(b) D_y
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
1000 / T (K^{-1})
M1 (fine-tuned)
M0 (MACE-MP-0-small)
saddles ±0.05 eV (judgment-based band)
prefactor fixed at 10^{13} s^{-1}
kMC, classical (T3-02)
kMC, saddles −0.05 eV
kMC, saddles +0.05 eV
MLIP-based harmonic-TST estimates; lattice model with 1 ML collective barriers; not DFT-validated; valid only above ~150–200 K